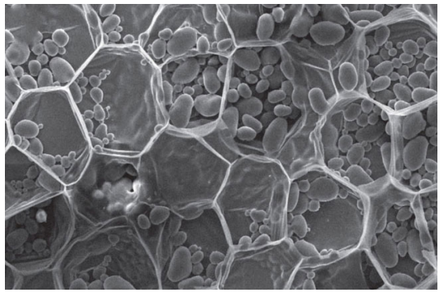


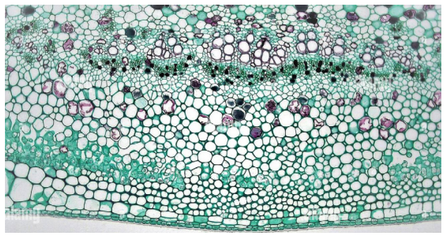
Cushioning

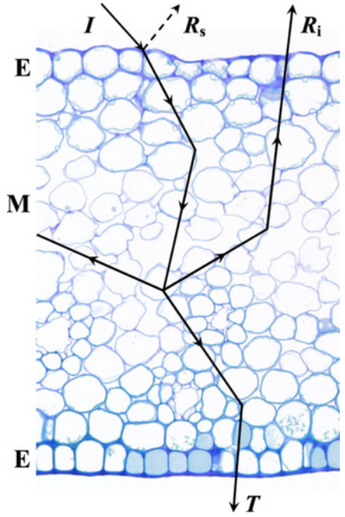
USING FLOWERS AS INSPIRATION

Biomimicry Research

PETAL MESOPHYLL





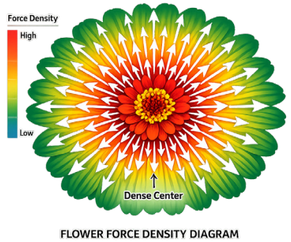


Petal mesophyll is a spongy, porous tissue with intercellular air spaces

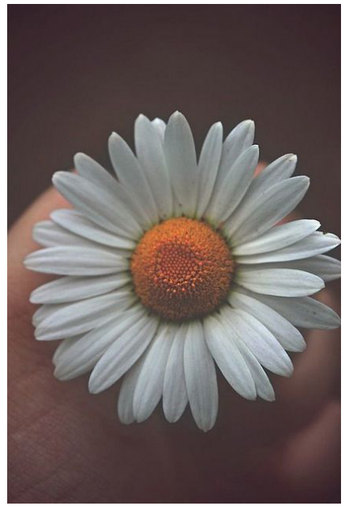
- Cellular structure → energy absorption via buckling
- Air gaps → compress first → dissipate force
- Cell walls → prevent total collapse

RADIAL GEOMETRY





FLOWER FORCE DENSITY DIAGRAM



Flowers use radial symmetry to distribute loads evenly

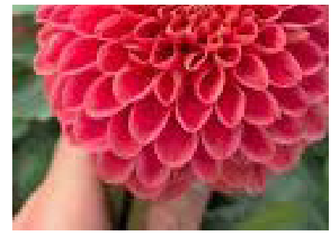
- Structural elements radiate outward from a center
- Multi-directional force pathways

PETAL LAYERING









Petals are arranged in overlapping layers with varying thickness and stiffness

- Outer petals → more flexible
- Inner petals → more constrained and dense
- A hierarchical structure

Lotus Flowers

Biological Observations

- The lotus flower uses **layered petal structures** arranged radially around a central core
- Petals have **overlapping geometry**, creating depth and structural redundancy
- Natural curvature allows petals to **flex and deform** under environmental forces
- The flower **maintains stability** while floating lightly on water





Design Direction

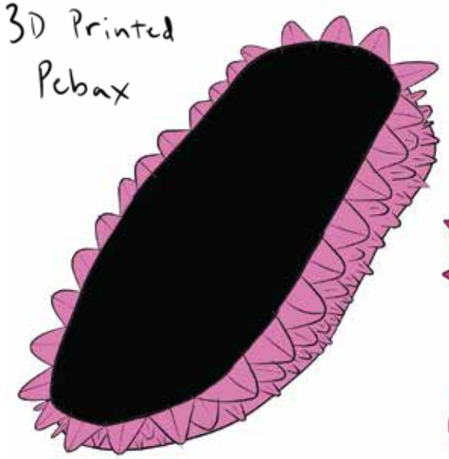
How might we design a midsole

that uses layered, petal-like structures

to distribute impact and create

a soft, stable underfoot experience?

3D Printed
Pebax



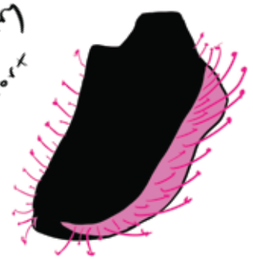


3D Printed
Pebax



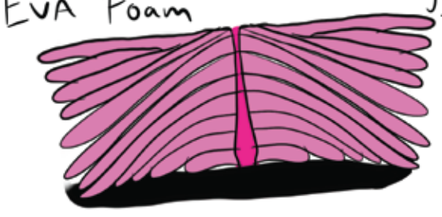


Replicating Soft Pollen Geometry
Interact directly w/ foot for comfort



UPSide down
flower layers
* Springy *

EVA Foam



Blossoming flower
Closed at heel for stability



3D Printed
Pebax

3D Printed
Pebax

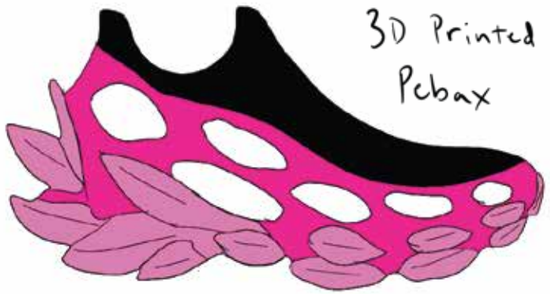


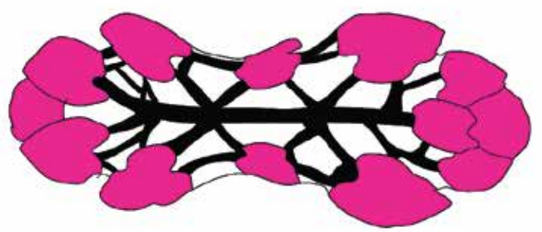
Bleeding Heart
flower Geometry



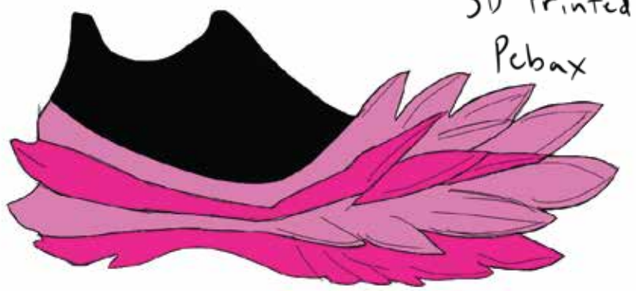
EVA Foam

3D Printed
Pebax





3D Printed
Pebax





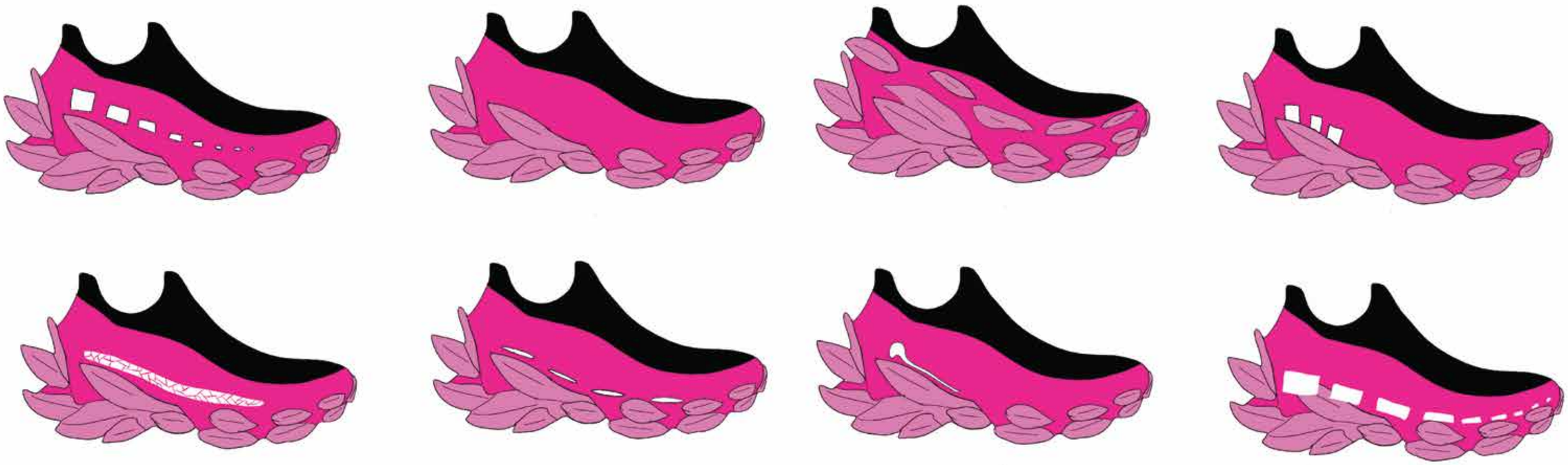


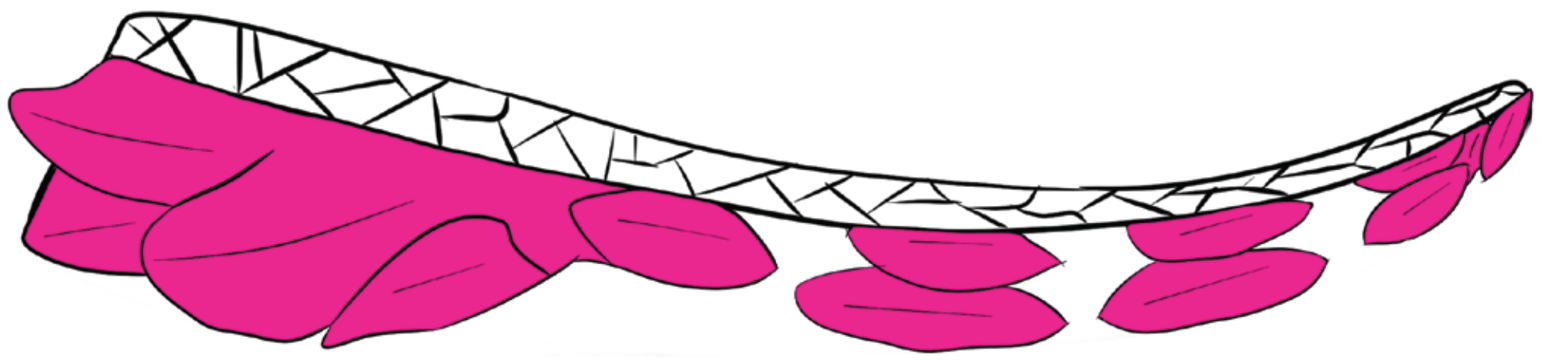
Petals curl from
Outsole into upper
Midsole ring

3D Printed
Pebax



	ELLIOT HILDNER	HW 9	DATE 4/26/26	PAGE 5
	SPORTS PRODUCT DESIGN	CUSHIONING SKETCHES	SCALE	

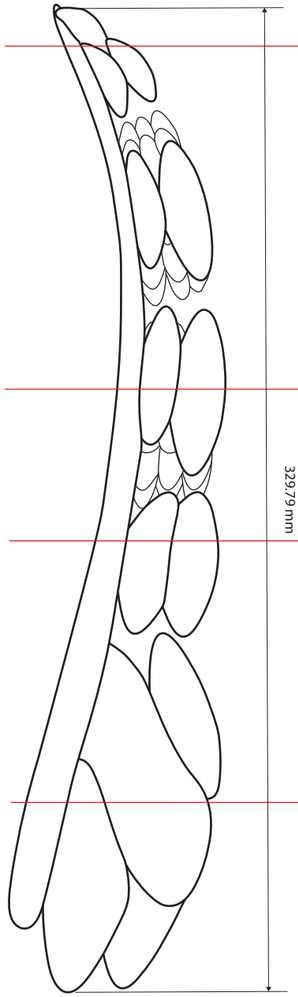




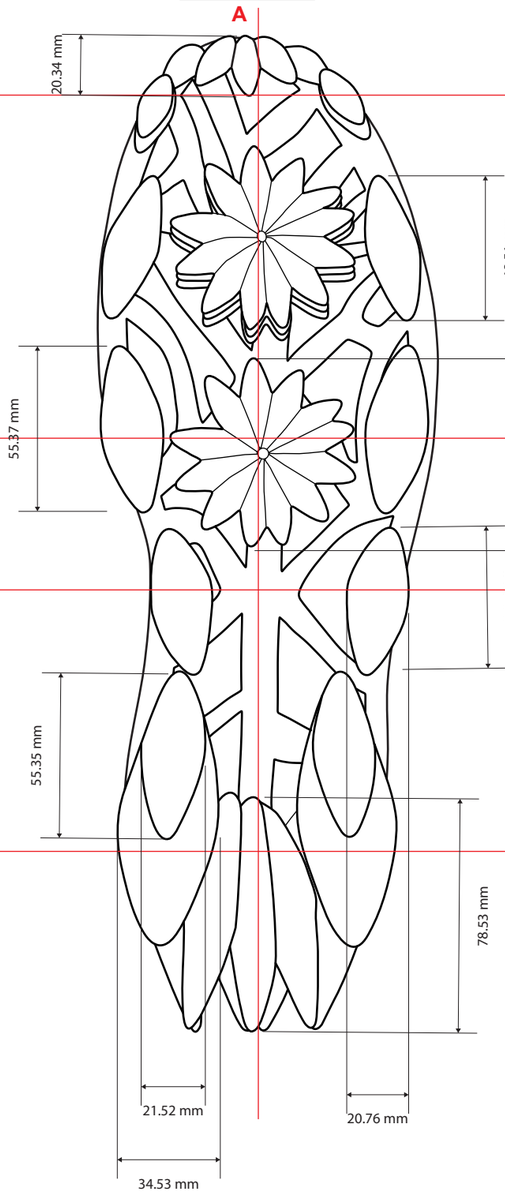


	ELLIOT HILDNER	HW 9	DATE 4/26/26	PAGE 6
	SPORTS PRODUCT DESIGN	CUSHIONING SKETCHES	SCALE	

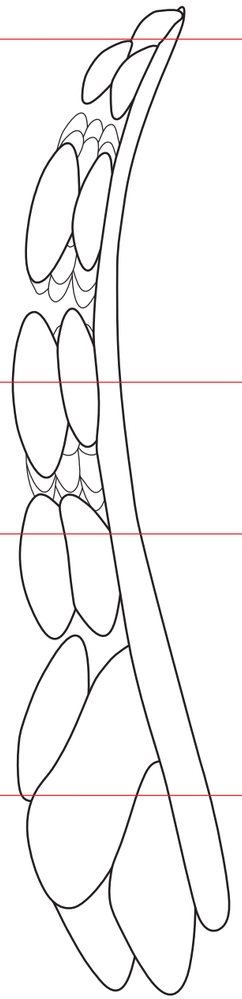
LATERAL VIEW



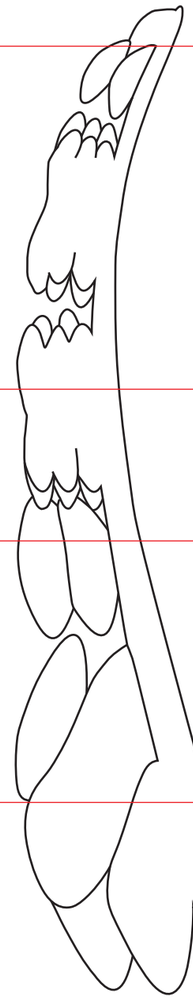
BOTTOM VIEW



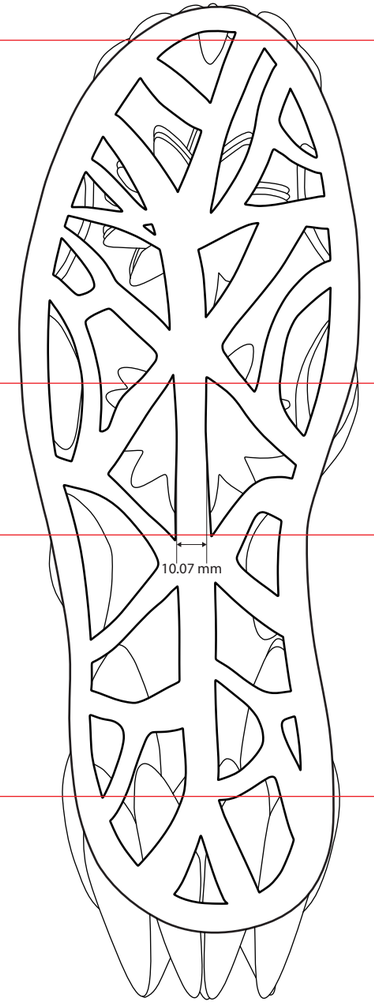
MEDIAL VIEW



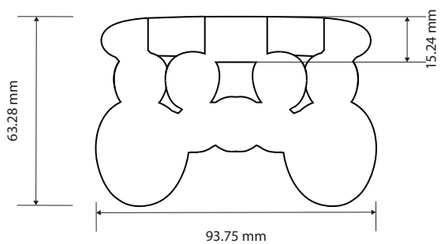
SECTION A-A



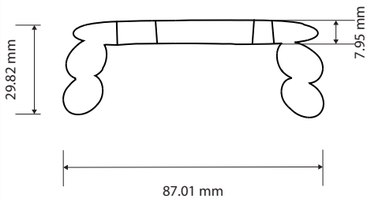
TOP VIEW



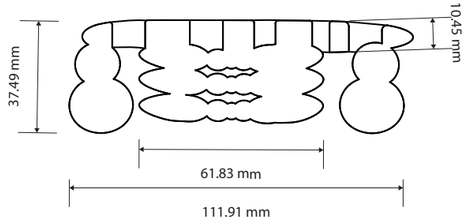
SECTION E-E



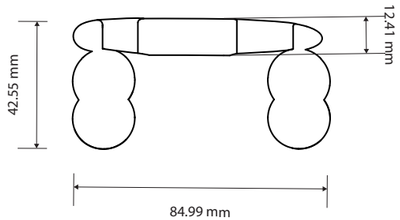
SECTION B-B



SECTION C-C



SECTION D-D



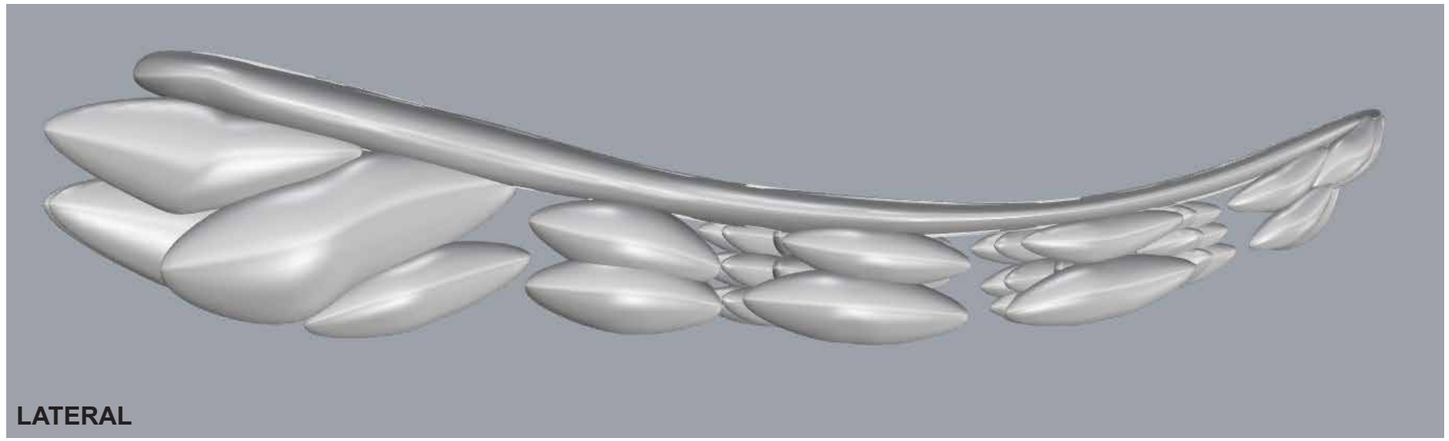
	ELLIOT HILDNER	HW10	DATE 5/1/26	UNITS mm
	SPORTS PRODUCT DESIGN	FLOWER CUSHION MIDSOLE	SCALE 1:2	PAGE 7



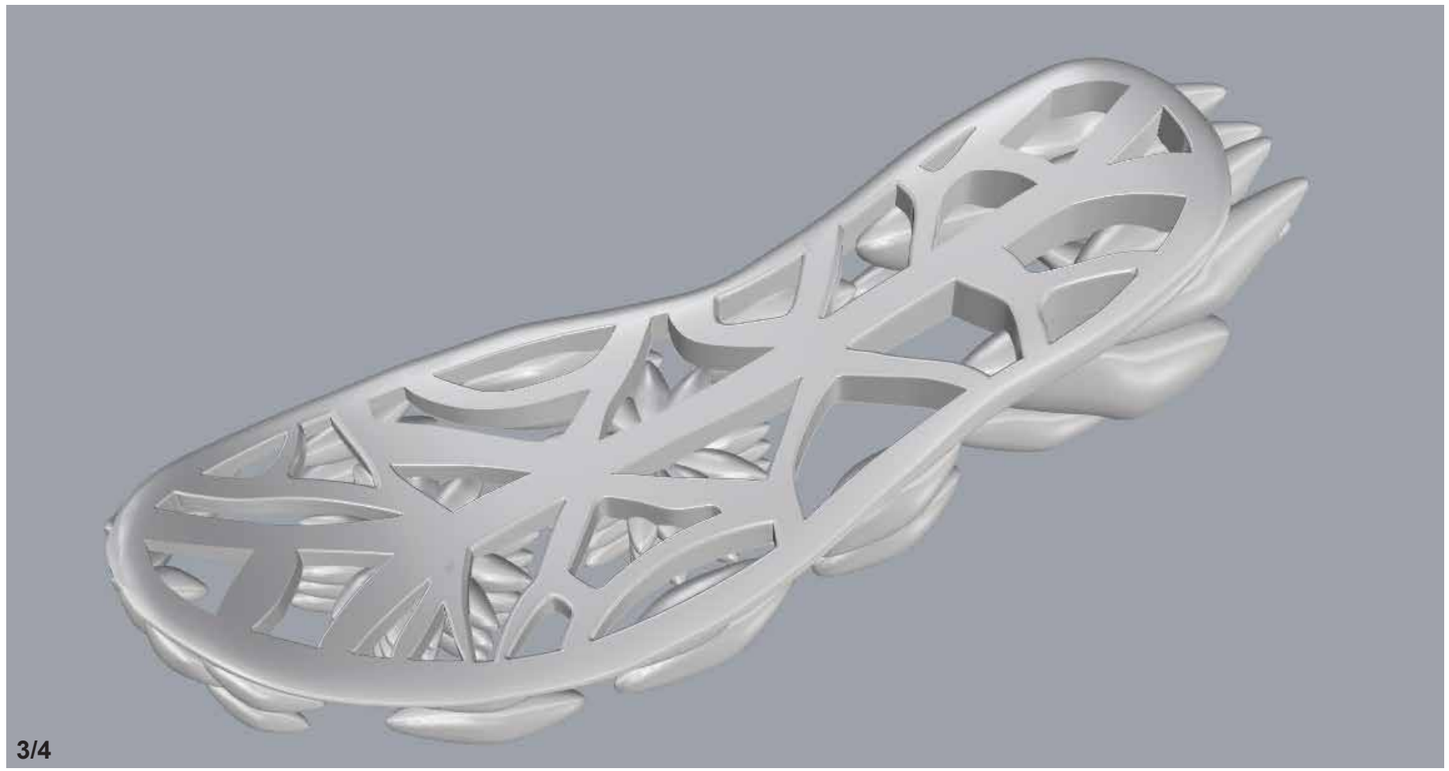
TOP



BOTTOM



LATERAL



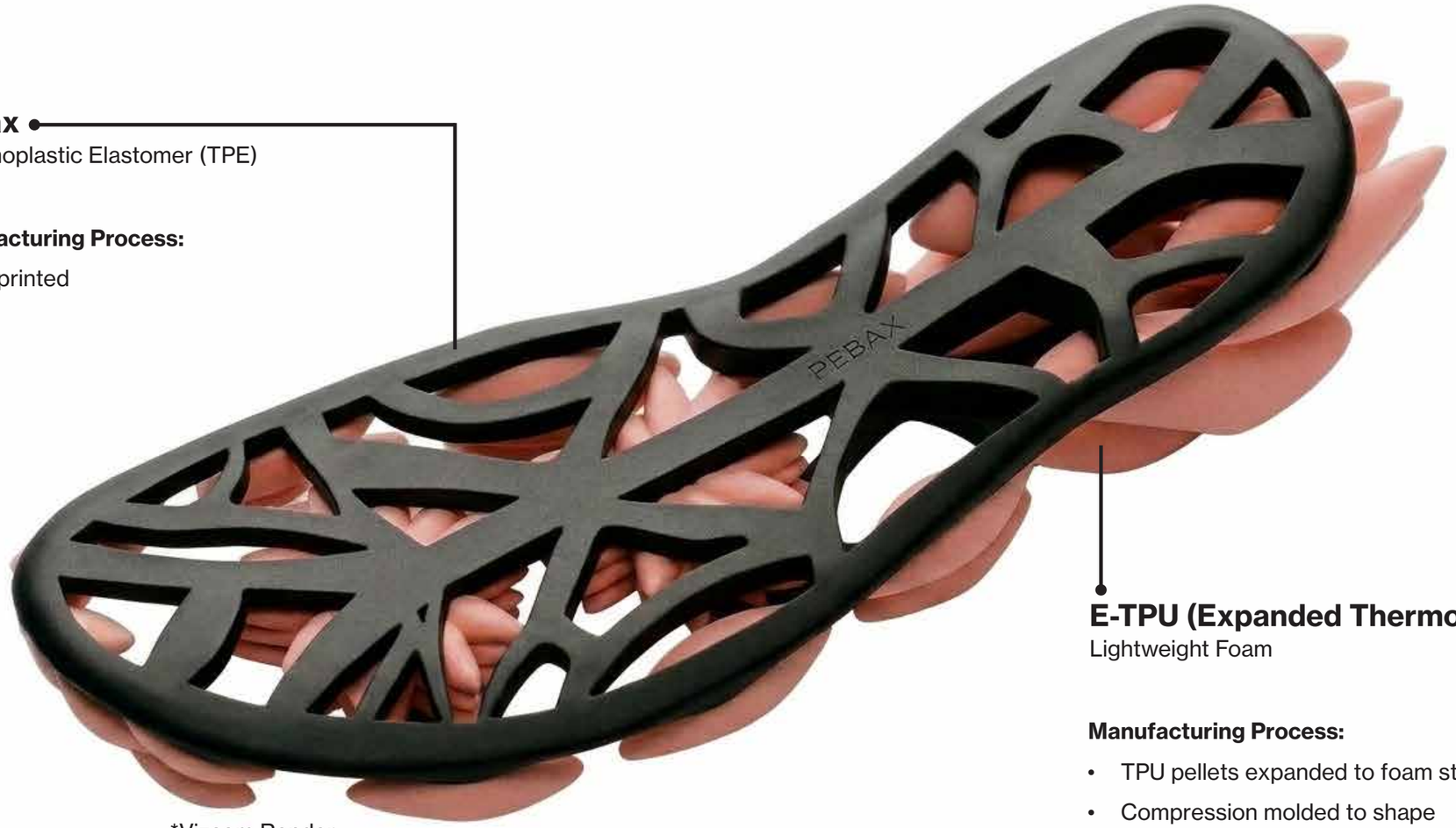
3/4

	ELLIOT HILDNER	HW 10	DATE 5/1/26	UNITS mm
	SPORTS PRODUCT DESIGN	CUSHION 3D CAD	SCALE 1:2	PAGE 8

Pebax
Thermoplastic Elastomer (TPE)

Manufacturing Process:

- 3D printed



*Vizcom Render

E-TPU (Expanded Thermoplastic Polyurethane)
Lightweight Foam

Manufacturing Process:

- TPU pellets expanded to foam structure
- Compression molded to shape

	ELLIOT HILDNER	HW 10	DATE 5/1/26	UNITS mm
	SPORTS PRODUCT DESIGN	MATS + MANUF	SCALE	PAGE 9

Lotus I

LIKE YOU'RE FLOATING ON WATER



*Vizcom Render

FEATURES & BENEFITS

- Layered, petal-like geometry for multi-stage compression to maintain support
- Radial arrangement on high pressure areas for balance and comfort
- Pebax web structure top layer for flexibility and added support

	ELLIOT HILDNER	HW 10	DATE 5/1/26	UNITS mm
	SPORTS PRODUCT DESIGN	3D RENDER	SCALE	PAGE 10

References

Chen, C., Airoldi, C. A., Lugo, C. A., Bay, R. K., Glover, B. J., & Crosby, A. J. (2021). Flower inspiration: Broad-angle structural color through tunable hierarchical wrinkles in thin film multilayers. Advanced Functional Materials, 31. https://doi.org/10.1002/adfm.202006256

De Paola, L., Veldhuis, T. A., Kraaij, M., Stavenga, D. G., Tiedge, K. J., & van der Kooi, C. J. (2025). Stacked scattering: The key to bright flowers lies in the mesophyll. American Journal of Botany. https://doi.org/10.1002/ajb2.70104

Hileman, L. C. (2022). Floral symmetry: The geometry of plant reproduction. Emerging Topics in Life Sciences, 6(3), 259–266. https://doi.org/10.1042/ETLS20210260

Niklas, K. J., & Spatz, H.-C. (2025). Plant biomechanics and the functional design of biological materials. Integrative and Comparative Biology, 65(1), 81–96. https://doi.org/10.1093/icb/icaeXXX

van der Kooi, C. J., Elzenga, J. T. M., Staal, M., & Stavenga, D. G. (2021). How to colour a flower: On the optical principles of flower coloration. Frontiers in Plant Science, 12, 745507. https://doi.org/10.3389/fpls.2021.745507